

Exchange Coupling Determines Metal-Dependent Efficiency for Iron- and Cobalt-Catalyzed Photochemical CO₂ Reduction

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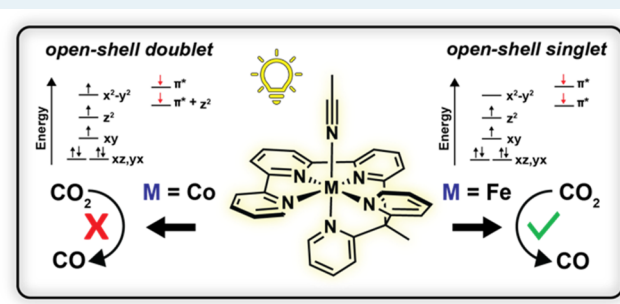
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ABSTRACT: Catalysts promoting multielectron charge delocalization offer selectivity for the CO₂ reduction reaction (CO₂RR) over the competing hydrogen evolution reaction. Here, we show metal–ligand exchange coupling as an example of charge delocalization that can determine the efficiency for photocatalytic CO₂RR. A comparative evaluation of iron and cobalt complexes supported by the redox-active ligand tpyPY2Me establishes that the two-electron reduction of [Co(tpyPY2Me)]²⁺ ([Co]²⁺) occurs at potentials 770 mV more negative than the [Fe(tpyPY2Me)]²⁺ ([Fe]²⁺) analogue by maximizing the exchange coupling in the latter compound. The positive shift in the reduction potential promoted by metal–ligand exchange coupling drives [Fe]²⁺ to be among the most active and selective molecular catalysts for photochemical CO₂RR reported to date, maintaining up to 99% CO product selectivity with total turnover numbers (TONs) and initial turnover frequencies exceeding 30,000 and 900 min^{−1}, respectively. In contrast, [Co]²⁺ shows much lower CO₂RR activity, reaching only ca. 600 TON at 83% CO product selectivity under similar conditions accompanied by rapid catalyst decomposition. The spin density plots of the two-electron reduced [Co]⁰ complex implicate a paramagnetic open-shell doublet ground state compared to the diamagnetic open-shell singlet ground state of reduced [Fe]⁰, rationalizing the observed negative shift in two-electron reduction potentials from the [M]²⁺ species and lowered CO₂RR efficiency for the cobalt complex relative to its iron congener. This work emphasizes the contributions of multielectron metal–ligand exchange coupling in promoting effective CO₂RR and provides a starting point for the broader incorporation of this strategy in catalyst design.

KEYWORDS: CO₂ reduction, exchange coupling, redox-active ligand, photocatalysis, electrocatalysis



INTRODUCTION

The selective and efficient transformation of carbon dioxide into value-added products offers a sustainable approach to generate fuels, foods, materials, and medicines from renewable feedstocks.^{1–4} Artificial photosynthesis is a potentially powerful strategy toward achieving this goal,^{5–15} where concept transfer from natural photosynthesis can be incorporated into synthetic systems capable of reducing CO₂ through light-initiated proton- and electron-transfer reactions. In this context, homogeneous artificial photosynthesis platforms offer the capacity to tune system performance using well-defined components, including the molecular catalyst, photosensitizer with an energy-matched reduction potential, and sacrificial electron donor with maximum quenching efficiency.^{11,16–22} Indeed, recent advances in molecular photocatalysis for the carbon dioxide reduction reaction (CO₂RR) include metal-dependent enhancement of organic photosensitizers,^{20,23} the use of hydrogen bonding,²⁴ electrostatic,^{14,25,26} or covalent²⁷ interactions between the photo-

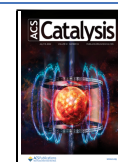
sensitizer and catalyst to improve electron transfer, and additives to increase CO₂ solubility.^{28–32}

In terms of a new catalyst design, first-row transition metal complexes supported by polypyridyl ligand frameworks^{30,32–52} have emerged as privileged scaffolds for CO₂RR. For example, we have recently found that strong metal–ligand exchange coupling between an iron center and a pentadentate polypyridyl ligand (tpyPY2Me) promotes its facile two-electron reduction to yield a reduced diamagnetic complex, [Fe(tpyPY2Me)] ([Fe]⁰), which we assigned to an open-shell singlet ground state that is composed of an intermediate-spin Fe²⁺ center antiferromagnetically coupled to a doubly reduced triplet tpyPY2Me ligand.⁴⁴ The metal–ligand cooperativity in

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this Fe^{2+} complex leads to a 640 mV positive shift in the first two ligand-centered reductions relative to the Zn^{2+} analogue. This feature enables catalysis of electrochemical CO_2RR at low overpotentials with high selectivity by delocalizing the electron density beyond the primary metal center. Moreover, the analogous nickel tpyPY2Me complex, $[\text{Ni}(\text{tpyPY2Me})]^{2+}$ ($[\text{Ni}]^{2+}$), has found application as an efficient redox mediator for controlling radical pathways by ligand-centered redox reactivity.⁵³

Against this backdrop, we sought to expand the reactivity of metal tpyPY2Me complexes to photochemical CO_2RR . Considering the precedent for cobalt-based CO_2RR catalysts,^{11,18,23,39,54–58} we now report a comparative study between iron and cobalt tpyPY2Me complexes for photochemical CO_2RR , including synthesis and characterization of the novel cobalt tpyPY2Me complex, $[\text{Co}(\text{tpyPY2Me})]^{2+}$ ($[\text{Co}]^{2+}$). We find that electron count makes a substantial contribution to metal–ligand cooperativity and redox behavior in these transition metal tpyPY2Me complexes. In contrast to the $[\text{Fe}]^{2+}$ complex, where the first two single-electron, ligand-based reduction waves are virtually superimposable, electrochemical characterization of $[\text{Co}]^{2+}$ shows a 1.39 V separation between the first and second redox processes where the two-electron reduction potential of $[\text{Co}]^{2+}$ is shifted negatively by 770 mV compared to $[\text{Fe}]^{2+}$. This marked difference in multielectron redox behavior correlates with the observed photocatalytic CO_2RR reactivity. Indeed, the $[\text{Fe}]^{2+}$ complex is compatible with photosensitizers spanning reduction potentials over a range of 1 V and can promote efficient light-driven CO_2 reduction with high activity and selectivity, achieving up to 99% CO product selectivity with total turnover numbers (TON) and initial turnover frequencies (TOFs) exceeding 30,000 and 900 min^{-1} , respectively, whereas the $[\text{Co}]^{2+}$ complex shows much lower CO_2RR activity under similar conditions (ca. 600 TON, 83% CO product selectivity). Density functional theory (DFT) calculations support these experimental findings by suggesting that the first reduction of $[\text{Co}]^{2+}$ to $[\text{Co}]^+$ is metal-centered and that the second reduction to $[\text{Co}]^0$ is accompanied by a significant rearrangement of electrons that enforces exchange coupling between a formal Co^{2+} center and a two-electron reduced ligand, resulting in a paramagnetic open-shell doublet. This electronic structure differs from the two-electron reduction of $[\text{Fe}]^{2+}$, resulting in a diamagnetic, open-shell singlet $[\text{Fe}]^0$ complex between a formal Fe^{2+} center and a two-electron reduced ligand. As such, we attribute the superior catalytic performance of the iron system over the cobalt congener to the more stable electronic configuration for reduced $[\text{Fe}]^0$ versus $[\text{Co}]^0$ driven by this metal–ligand exchange coupling. Taken together, these data highlight the importance of metal–ligand redox cooperativity in developing efficient CO_2RR catalysts, which can be carried forward as a design principle for a broader range of multielectron redox transformations of interest.

RESULTS AND DISCUSSION

Synthesis and Structural Characterization of the $[\text{Co}(\text{tpyPY2Me})]^{2+}$ Complex. The $[\text{Co}(\text{tpyPY2Me})]^{2+}$ complex ($[\text{Co}]^{2+}$) was obtained in a nearly quantitative yield by stirring the ligand tpyPY2Me with an equimolar amount of the cobalt precursor $[\text{Co}(\text{OTf})_2(\text{CH}_3\text{CN})_2]$ in dry CH_3CN at room temperature for 12 h (Figure 1a). Diffraction quality crystals for $[\text{Co}]^{2+}$ were obtained by slow vapor diffusion of diethyl ether into a saturated CH_3CN solution. The crystal

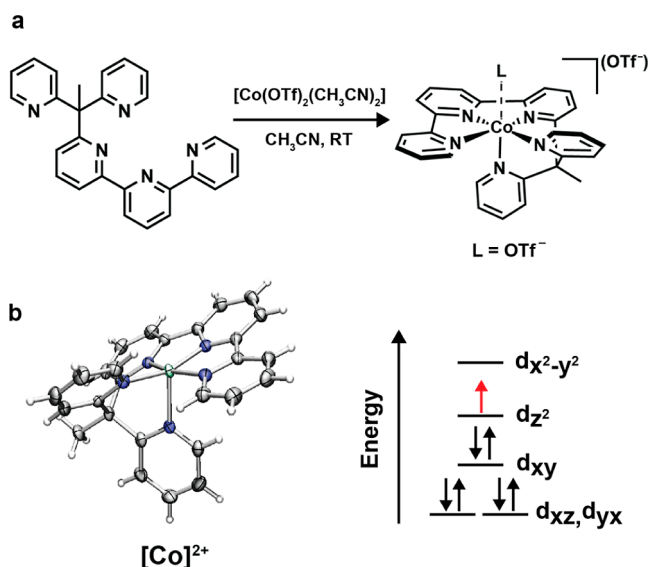


Figure 1. (a) Synthetic scheme and (b) X-ray crystal structure for the $[\text{Co}(\text{tpyPY2Me})]^{2+}$ ($[\text{Co}]^{2+}$) complex with its simplified crystal field splitting diagram. Thermal ellipsoids are plotted at 80% probability.

structure of $[\text{Co}]^{2+}$ (Figure 1b) shows its connectivity as a five-coordinate square pyramidal complex analogous to the previously reported $[\text{Zn}(\text{tpyPY2Me})]^{2+}$ analogue.⁴⁴ The $\text{Co}-\text{N}_{\text{tpy}}$ bond lengths range from 1.8975(17) to 1.9610(18) Å, consistent with a low-spin Co^{2+} configuration, exhibiting shorter bond lengths compared to high-spin Co d^7 complexes with a similar coordination environment.^{59,60} Analysis of the two $\text{C}_{\text{py}}-\text{C}_{\text{py}}$ bonds within the terpyridine moiety and the $\text{C}-\text{N}$ bonds of the terpyridine chelate rings (with average bond lengths of 1.471 and 1.356 Å, respectively) suggest that the ground state of $[\text{Co}]^{2+}$ bears a neutral tpyPY2Me ligand and a Co^{2+} metal center;⁶¹ this assignment was supported by an effective magnetic moment of 1.8 μB determined by the Evans' method (Figures 1b and S2).

Electrochemical Characterization Reveals Differences in Redox Behavior, Catalytic Efficiency, and Reductive Stability between $[\text{Co}(\text{tpyPY2Me})]^{2+}$ and $[\text{Fe}(\text{tpyPY2Me})]^{2+}$ for CO_2RR . As a starting point to compare the redox properties of the $[\text{Co}]^{2+}$ and $[\text{Fe}]^{2+}$ complexes, we measured the cyclic voltammogram (CV) of $[\text{Co}]^{2+}$ under an argon atmosphere. The CV shows a quasi-reversible wave at $E_{1/2} = +0.023$ V versus $\text{Fc}^{+/0}$ and three reversible waves at $E_{1/2} = -0.81, -2.20, -2.34$ V versus $\text{Fc}^{+/0}$ (Figure 2). The currents for all redox waves vary linearly with the square root of the scan rate, indicative of diffusion-controlled processes (Figure S1). The first wave at $E_{1/2} = +0.02$ V versus $\text{Fc}^{+/0}$ is assigned to a $\text{Co}^{2+/3+}$ oxidation, as its quasi-reversible nature is characteristic for cobalt complexes exhibiting low/high spin transitions coupled to a redox process.^{62–65} The second wave is assigned to a $\text{Co}^{2+/+}$ reduction process, which is positively shifted by +0.39 V compared to the Co^{2+} complex of bpy2PYMe , and consistent with an observed pattern of stabilization with increasing conjugation in pyridine-based redox-active ligands.⁶⁵ We assign the latter two waves at $E_{1/2} = -2.20$ and -2.34 V versus $\text{Fc}^{+/0}$ as ligand-centered reductions by comparison to the $[\text{Zn}(\text{tpyPY2Me})]^{2+}$ analogue bearing a redox-silent Zn^{2+} center, whose first two ligand-centered redox processes lie between -1.56 and -2.41 V versus $\text{Fc}^{+/0}$.⁴⁴ This electrochemical behavior contrasts with the two closely spaced ligand-

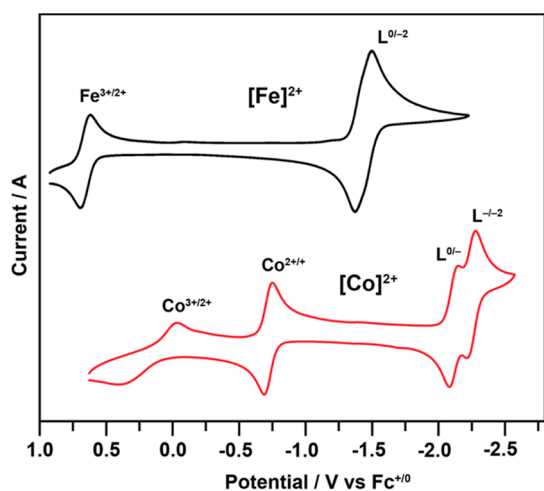


Figure 2. CVs of $[\text{Fe}]^{2+}$ (black) and $[\text{Co}]^{2+}$ (red). Conditions: 1 mM $[\text{M}(\text{tpyPY2Me})]^{2+}$ complex and 0.1 M NBu_4PF_6 in CH_3CN under an Ar atmosphere using a GC electrode.

dependent reductions in $[\text{Fe}]^{2+}$ centered at -1.43 V versus $\text{Fc}^{+/0}$ (Figure 2). As such, the change from iron to cobalt shifts the first two reductions in a negative direction by 770 mV, indicating a significant difference in the thermodynamic stability of metal–ligand electronic coupling in the reduced cobalt complex compared to its iron congener.

We next moved on to screen $[\text{Co}]^{2+}$ for electrochemical CO_2RR reactivity. Upon addition of CO_2 , we observe a current enhancement, which increases further upon the addition of 0.1 M phenol as an acid source (Figure 3a). Under an Ar atmosphere, the addition of 0.1 M phenol results in a larger relative current enhancement, indicative of hydrogen evolution reaction (HER) reactivity (Figure 3b). We then measured product selectivity and long-term catalyst stability via controlled potential electrolysis (CPE) experiments conducted at various applied potentials. Initially, we used glassy carbon (GC) as a working electrode for CPE experiments; however, the current density asymptotically decays within the first 50 s at the selected potentials (Figure S3a). CVs recorded before and immediately after electrolysis show a total loss in current response; upon polishing the GC electrode after electrolysis and re-submerging in the same electrolysis cell, the voltammograms show recovery of the same pre-electrolysis wave (Figure

S3c). This behavior is indicative of catalyst deposition onto the working electrode, resulting in an inactive film that limits charge transfer during the electrolysis experiment. To limit this deposition, we turned to the use of reticulated vitreous carbon (RVC) foam as a carbon-based electrode with higher surface area for further electrolysis experiments. Use of RVC foam led to a modest enhancement in the stability and enabled more charge to be passed during the electrolysis experiment (Figure S3d,e). At an applied potential of -2.09 versus $\text{Fc}^{+/0}$ under a CO_2 atmosphere in the presence of 0.1 M phenol, $[\text{Co}]^{2+}$ achieved 64% Faradaic efficiency for CO formation (Table S1). During CPE experiments with the RVC working electrode, we observed a color change in the electrolyte that was not observed when the GC working electrode was used, where the pale orange-colored solution characteristic of $[\text{Co}]^{2+}$ converted to a dark red purple-colored solution. We speculate that this species is most likely the two-electron reduced species, $[\text{Co}]^0$, based on a similar color change observed for the chemical reduction of $[\text{Fe}]^{2+}$ to $[\text{Fe}]^0$ using decamethylcobaltocene.⁴⁴ We have not yet been successful in isolating the reduced cobalt species owing to its redox instability, but its appearance in bulk solution suggests that it contributes to the lower Faradaic efficiency and electrochemical stability observed for the cobalt catalyst relative to the iron analogue.

Evaluation of $[\text{Co}(\text{tpyPY2Me})]^{2+}$ as a Catalyst for Photochemical CO_2RR . Building upon these electrochemical results, we reasoned that moving to photochemical CO_2RR would enable lower catalyst loading (1 mM for electrolysis to 2 μM for photolysis, vide infra) to mitigate unproductive $[\text{Co}]^0$ accumulation by dispersing it in solution as opposed to concentrating the reduced species at the electrode double layer. Indeed, in a photocatalytic system, electron transfer operates via the bimolecular reaction of the catalyst with a photosensitizer; thus, we expected improved catalyst stability relative to the electrolysis experiments. As a starting point, we evaluated $[\text{Co}]^{2+}$ as a catalyst for photochemical CO_2RR . In a typical experiment, 2 μM $[\text{Co}]^{2+}$ was added to a CO_2 -saturated CH_3CN solution containing 200 μM photosensitizer, 50 mM triethylamine (TEA) as a sacrificial electron donor, and 1 M phenol as a proton source. Reactions were irradiated in 1 h intervals using a blue LED light source. Multiple photosensitizers were chosen for screening owing to their close energy match to reduction waves of $[\text{Co}]^{2+}$, including $[\text{Ru}(\text{bpy})_3]^{2+}$ (RuPS, $E^0 = -1.71$ V vs $\text{Fc}^{+/0}$), as well as the

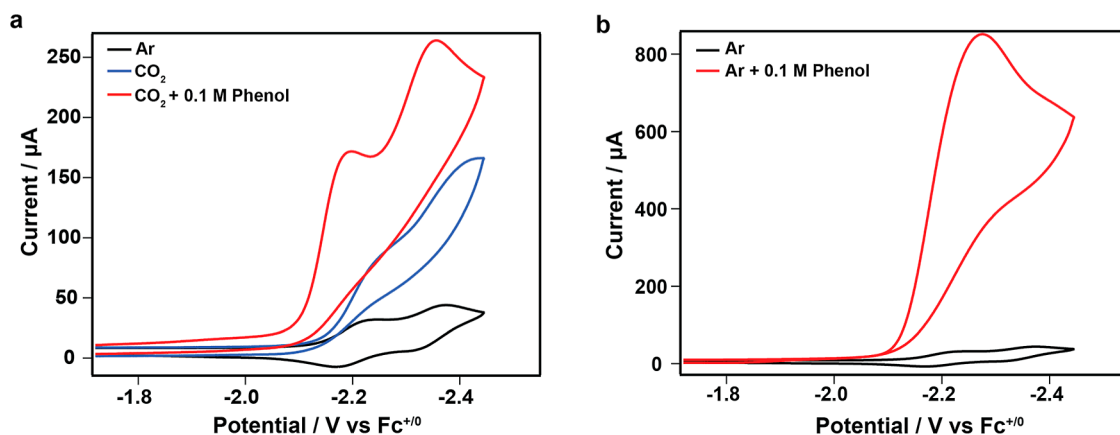


Figure 3. CVs showing catalytic responses upon addition of 0.1 M phenol to $[\text{Co}]^{2+}$ (a) under CO_2 or (b) Ar atmosphere. Conditions: 1 mM $[\text{Co}]^{2+}$ in 0.1 M $\text{NBu}_4\text{PF}_6/\text{CH}_3\text{CN}$ using a GC electrode.

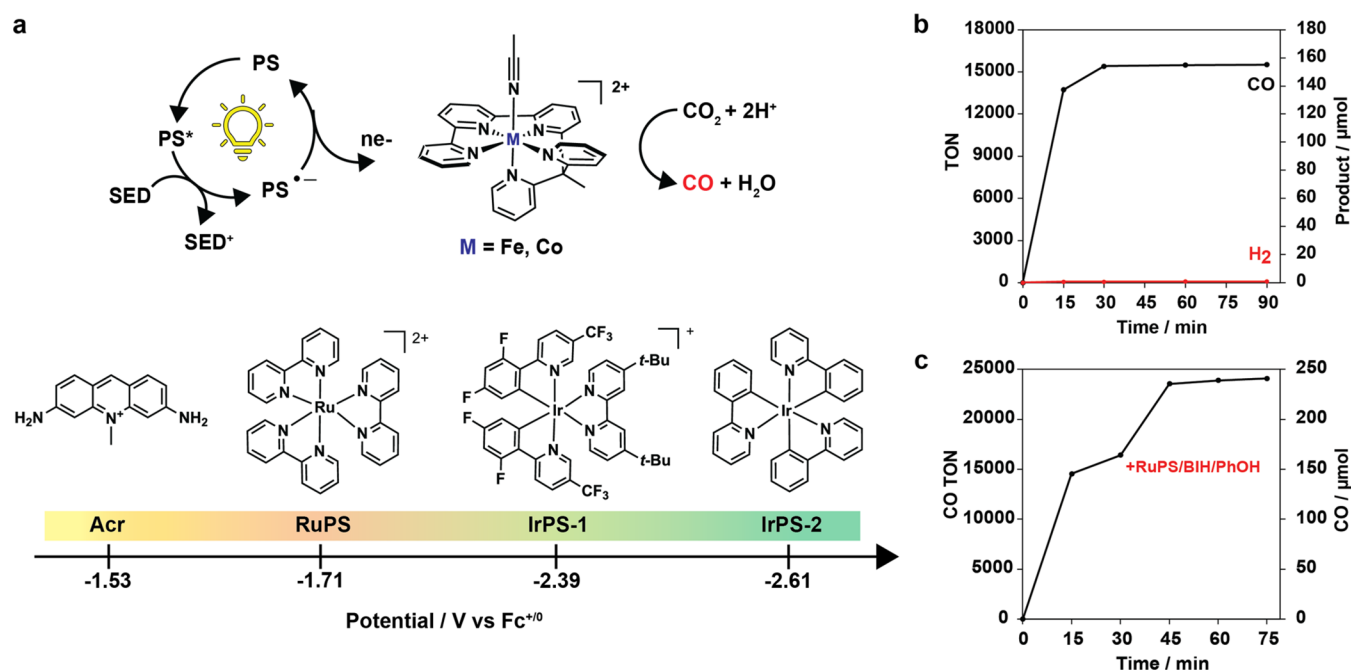


Figure 4. (a) Photochemical CO₂RR catalyzed by iron and cobalt tpyPy₂Me complexes, [Fe]²⁺ and [Co]²⁺, using Acr, RuPS, IrPS-1, and IrPS-2 as photosensitizers with increasing driving forces for reduction. (b) Photocatalytic activity of [Fe]²⁺ over 90 min and (c) with replenishing the photosensitizer, sacrificial electron donor, and proton source every 30 min. Conditions: 2 μM [Fe]²⁺, 200 μM RuPS, 100 mM BIH, and 1 M Phenol in CO₂-saturated CH₃CN solution.

iridium complexes [Ir(dFCF₃ppy)₂(dtbbpy)]⁺ (**IrPS-1**, $E^0 = -2.39$ V vs Fc^{+/0}) and Ir(ppy)₃ (**IrPS-2**, $E^0 = -2.61$ V vs Fc^{+/0}) (Figure 4a). The results of photochemical CO₂RR screening studies show some improvements and similarities in activity trends compared to electrochemical CO₂RR experiments (Figure S4 and Table S2). The average activity of the **IrPS-1** system over a 15 h period shows that in the first hour, CO₂ reduction is 93% selective for CO with an average TON of 120. Data collected at subsequent time points show a marked drop in activity as well as a loss of CO₂RR selectivity down to 50–57% CO product due to increased HER. We hypothesized that the modest observed CO₂RR activity may be a consequence of inefficiencies in the photocatalytic system design, in particular the requirement for a high concentration of acid that may participate in hydrogen bonding with the basic TEA sacrificial electron donor, which can possibly lower its quenching efficiency. We turned to 1,3-dimethyl-2-phenyl-2,3-dihydro-1*H*-benzo[*d*]imidazole (BIH) as a replacement for TEA, which has literature precedent as an excellent quencher for photocatalytic CO₂ reduction using organic, ruthenium-, and iridium-based photosensitizers.^{20,39,45,66,67} With 100 mM BIH, [Co]²⁺ exhibits a switch in selectivity from CO₂RR to HER using either RuPS or IrPS-1 as a photosensitizer, with an average H₂ TON of 488 achieved in 1 hour. Lowering the BIH concentration to 50 mM improves the selectivity to 67–83% CO, with **IrPS-1** giving the highest TON of 647 in 1 hour of photolysis. Despite the observed gains in activity and selectivity with lower BIH loading, reactions still show a loss in CO₂RR selectivity beyond the first hour, analogous to results obtained with TEA, indicating that lowering BIH loading only slowed off-pathway H₂ evolution and deactivation pathways rather than eliminating them. Indeed, dynamic light scattering (DLS) experiments confirm that within 15–30 min of irradiation, [Co]²⁺ decomposes into a particulate species with sizes in the range of 100 to 1000 nm under catalytic conditions (Figure

S5a), showing that photochemical CO₂RR reactivity is ultimately limited by the stability of the catalyst under reductive conditions.

[Fe(tpyPY₂Me)]²⁺ Shows Superior Catalytic Activity to [Co(tpyPY₂Me)]²⁺ for Photochemical CO₂RR. We then turned our attention to [Fe]²⁺ as a catalyst for photochemical CO₂RR, anticipating that it would exhibit better activity, selectivity, and stability compared to [Co]²⁺ considering its two-electron reduction at more positive potentials and superior performance as an electrochemical CO₂RR catalyst. Optimization of photocatalysis in CH₃CN solution with 1 M phenol as a proton source led us to screen a range of organic and noble-metal photosensitizers compatible under these conditions. These photosensitizers include the mildly reducing organic dye acriflavine (**Acr**, $E^0 = -1.53$ V vs Fc^{+/0}), whose reduction potential lies just 100 mV more negative than the catalytically relevant [Fe]²⁺ reduction, as well as RuPS and the highly reducing **IrPS-1** and **IrPS-2** sensitizers discussed previously (Figure 4a). In a typical experiment, 2 μM [Fe]²⁺ catalyst was added to a CO₂-saturated CH₃CN solution containing 200 μM photosensitizer, 100 mM BIH as a sacrificial electron donor, and 1 M phenol as a proton source. The reactions were irradiated in either 15- or 30 min intervals using a blue LED light source. We first examined photocatalysis using the RuPS/BIH system; a TOF of 916 min⁻¹ is reached within the first 15 min of the reaction, which to the best of our knowledge is the highest reported TOF value for CO₂RR driven by a molecular catalyst.²⁵ A plateau is reached after 30 min (Figure 4b), resulting in a total TON of 15,520 with 99% selectivity for the CO product with negligible H₂ evolution. During this time, we observe bleaching of the photosensitizer from a bright orange to pale yellow color, indicating degradation of RuPS. This degradation has been reported to proceed by dissociation of bipyridine ligands from the Ru center during light irradiation.^{65,68} We confirmed that

replenishing RuPS with BIH and phenol could partially recover the activity of $[\text{Fe}]^{2+}$ after the initial plateau, resulting in a maximum TON of 24,069 for CO production after a total of 75 min (Figure 4c). The photocatalytic activity could be further amplified by lowering the catalyst loading to 0.2 μM , achieving a TON of 30,349 within 45 min while still producing 30 μmol of CO with 97% product selectivity (Table 1, entry

Table 1. Photocatalytic Activity of $[\text{Fe}]^{2+}$ under Various Conditions^a

entry	TON		product (μmol)		selectivity (%)
	CO	H ₂	CO	H ₂	
1	15520	86	155	0.86	99
2 ^b	30349	1013	30	1	97
3 ^c	43	52	0.4	0.5	52
4 ^d	112	0	1.1	0	100
5 ^e	150	0	1.5	0	100
6 ^f	6	0	0	0	100
7 ^g	0	222	0	2.2	0
8 ^h	12749	163	127	1.6	98
9 ⁱ	28712	6527	28	6.5	81
10 ^j	18502	141	185	1.4	99
11 ^k	6710	0	67	0	100

^aReaction conditions: standard conditions using 2 μM $[\text{Fe}]^{2+}$ catalyst, 200 μM RuPS photosensitizer, 100 mM BIH quencher, and 1 M phenol as a proton source in 5 mL of CO_2 -saturated CH_3CN solution irradiated for 90–120 min unless otherwise noted. ^bUsing 0.2 μM catalyst. ^cWithout catalyst. ^dWithout RuPS. ^eWithout BIH. ^fWithout phenol. ^gUnder an Ar atmosphere. ^hUsing 200 μM IrPS-1 at 2 μM catalyst loading. ⁱUsing 200 μM IrPS-1 at 0.2 μM catalyst loading. ^jUsing 200 μM IrPS-2. ^kUsing 125 μM acriflavine (Acr).

2). Control experiments performed confirm that the catalyst, photosensitizer, sacrificial electron donor, proton source, and CO_2 are all essential for the reaction to proceed (Table 1, entries 4–7). Moreover, in contrast to the cobalt system, DLS measurements of the reaction mixture before and during peak catalytic activity at 0, 15, and 30 min confirm that the iron system is homogeneous, with no evidence of particle formation under photocatalytic conditions (Figure S5b).

We then employed Stern–Volmer analysis for the excited-state quenching of RuPS using BIH and $[\text{Fe}]^{2+}$ to probe the mechanism for photocatalysis (Figure 5). The bimolecular quenching rate constants (k_q) obtained for $[\text{Fe}]^{2+}$ ($2.7 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$) and BIH ($2.2 \times 10^9 \text{ M}^{-1} \text{ s}^{-1}$) are comparable, indicating that oxidative and reductive quenching mechanisms are both accessible, but we reason that the large excess of BIH used relative to $[\text{Fe}]^{2+}$ in the system will favor the reductive quenching pathway.⁶⁹ The quantum yield for CO production from the RuPS system with 2 μM $[\text{Fe}]^{2+}$ was determined to be $\Phi = 11.1\%$ by ferrioxalate actinometry (see the Supporting Information for details).

As such, we turned our attention to the water-soluble organic dye Acr, which has a similar redox potential to Purpurin and has been reported for photocatalytic water splitting and CO_2 reduction at 400 μM concentrations in $\text{CH}_3\text{CN}/\text{H}_2\text{O}$ mixtures.^{57,70} The maximum solubility of Acr in pure CH_3CN is approximately 125 μM . The use of a less reducing photosensitizer at lower concentrations led to slower photochemical CO_2RR catalysis, with a TON of 6,710 with 100% selectivity for the CO product obtained after 2 h of photolysis. We observe a gradual decrease in activity over time,

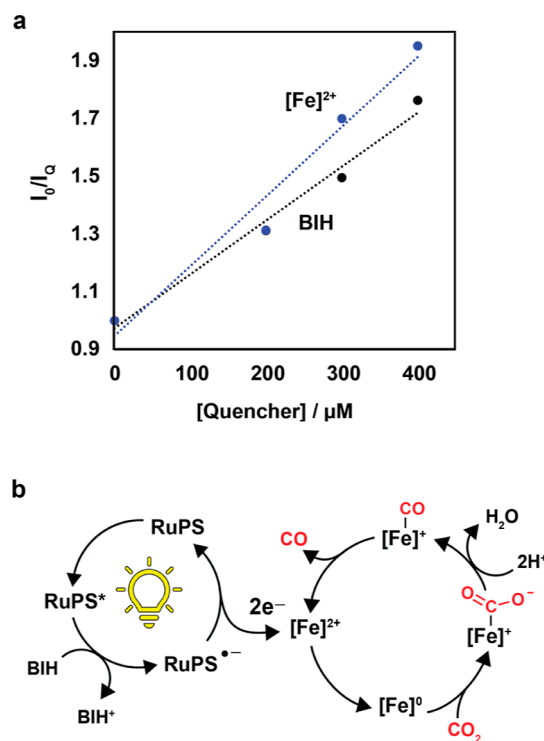


Figure 5. (a) Stern–Volmer plots for RuPS quenching with $[\text{Fe}]^{2+}$ and BIH. Conditions: 50 μM RuPS with BIH in a CO_2 -saturated CH_3CN solution. (b) Predicted reductive quenching mechanism for photocatalytic CO_2 reduction using $[\text{Fe}]^{2+}$ as the catalyst, BIH as the sacrificial electron donor, and RuPS as the photosensitizer.

likely due to the decomposition of the photosensitizer, as evidenced by bleaching observed much like with RuPS. These results establish a highly selective and active noble-metal free molecular system for catalytic photochemical CO_2RR using Acr and $[\text{Fe}]^{2+}$ components.

Finally, we examined $[\text{Fe}]^{2+}$ -catalyzed photochemical CO_2RR with the iridium-based photosensitizers IrPS-1 and IrPS-2 to provide a direct comparison with the results obtained using $[\text{Co}]^{2+}$. In addition, Ir–C cyclometallation in these complexes provides longer photostability compared to RuPS and Acr,⁷¹ providing an opportunity to probe contributions of photosensitizer stability to CO_2RR catalysis. Using IrPS-1 as a photosensitizer with 2 μM $[\text{Fe}]^{2+}$ as a CO_2RR catalyst results in a TON of 12,749 with 98% selectivity for CO (Table 1, entry 8), representing a 20-fold improvement over $[\text{Co}]^{2+}$. Lowering the catalyst concentration to 0.2 μM gives a 2-fold higher TON of 28,712 but with only 81% CO selectivity (Table 1, entry 9). Using IrPS-2 drives the highest TON of 18,502 at 2 μM $[\text{Fe}]^{2+}$ catalyst loading conditions (Table 1, entry 10). Interestingly, the plateau in catalytic activity observed using IrPS-2 shows distinct behavior compared with the one observed with RuPS since the former photosensitizer is cyclometalated and should not be the source of system deactivation. Indeed, we found that for the IrPS-2 system, introduction of additional $[\text{Fe}]^{2+}$, but not BIH and phenol, can enable recovery of activity (Figure S6), in contrast to the RuPS system where replenishing with RuPS/BIH/phenol can restore catalysis (Figure 4c). These results suggest that using the more reducing iridium photosensitizers can drive rapid photochemical CO_2RR with high selectivity for CO production but also accelerate catalyst deactivation. The key results obtained for use of $[\text{Fe}]^{2+}$ as a photocatalyst using all

the four photosensitizers and control experiments are summarized in Table 1.

Electronic Structure Calculations for Reduced Iron, Cobalt, and Nickel tpyPY2Me Complexes Show Key Differences in Metal–Ligand Exchange Coupling That Correlate with the Observed Redox and Catalytic Stability. Computational predictions for the first and second reduction potentials of the $[\text{Co}]^{2+}$ complex, modeled as $[\text{Co}(\text{tpyPY2Me})(\text{CH}_3\text{CN})]^{2+}$, were obtained using DFT calculations employing $\omega\text{B97X-V}$ functionals and CH_3CN solvent. DFT analysis shows that the first reduction of $[\text{Co}]^{2+}$ to $[\text{Co}]^+$ is a metal-centered process (Figure 6, left), resulting

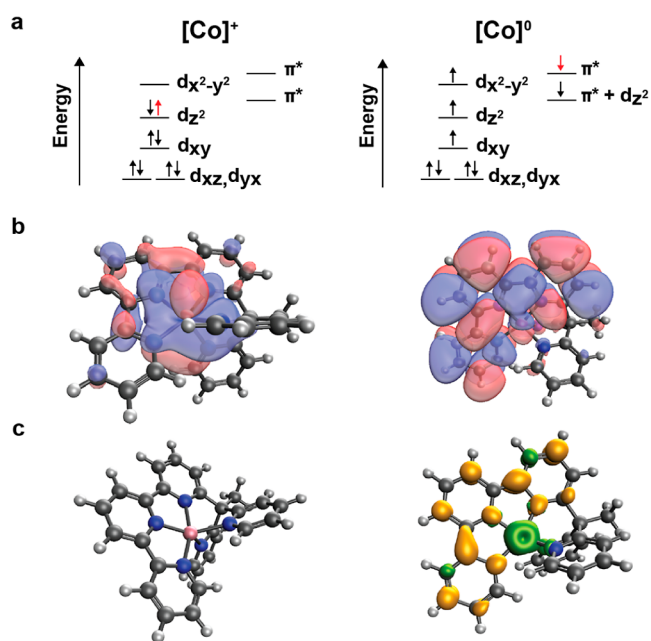


Figure 6. DFT analysis of electrochemically accessible reduced species, $[\text{Co}]^+$ and $[\text{Co}]^0$. For each reduced species, the following are shown: (a) qualitative molecular orbital diagram consistent with DFT calculations, with newly added electrons highlighted in red; (b) relevant computed MO; and (c) the computed spin density plot. The DFT results show that the first reduction is cobalt-centered to generate a formal Co^+ species, whereas the second reduction is ligand-centered, leading to a substantial rearrangement of electron density resulting in restoration of a formal Co^{2+} complex with a doubly reduced tpyPY2Me ligand.

in a singlet ground state ($S = 0$) and a predicted redox potential ($E^0 = -0.887$ V vs $\text{Fc}^{+/0}$) that is consistent with the experimental value (-0.81 V). Additional computational results from localized orbital bonding analysis (LOBA) confirm that the $[\text{Co}]^+$ complex can be described as a formal Co^+ center bound to a neutral tpyPY2Me ligand. The first reduction is accompanied by the loss of the axial solvent ligand, resulting in a five-coordinate $[\text{Co}]^+$ complex. DFT analysis of the doubly reduced $[\text{Co}]^0$ complex is consistent with the population of a ligand-based π^* orbital belonging to the terpyridine fragment of the ligand (Figure 6, right). The calculated redox potential ($E_0 = -2.22$ V vs $\text{Fc}^{+/0}$) matches the experimentally determined value ($E_{1/2} = -2.34$ V) and has a calculated spin state of $S = 1/2$. Interestingly, the spin density plot for $[\text{Co}]^0$ (Figure 6c, right) suggests that this second reduction is accompanied by significant rearrangement of electrons occupying the metal d orbitals, leading to the

population of a low-lying metal–ligand ($\pi^* + d_z^2$) orbital with concomitant reordering of Co-based electrons into a high-spin configuration. The computational results suggest that the $[\text{Co}]^0$ species exists as a formal Co^{2+} center bound to $(\text{tpyPY2Me}^{\bullet\bullet})^{2-}$. Taken together, these results highlight an intriguing outcome of formal oxidation of the metal center through reduction of the overall complex, made possible via delocalized reduction events as a result of metal–ligand exchange coupling. Coordinates of the optimized structures for $[\text{Co}]^+$ and $[\text{Co}]^0$ obtained using the $\omega\text{B97X-D}$ functional are shown in Tables S6 and S7.

The reduction pathway of $[\text{Co}]^{2+}$ contrasts with the computational studies of the analogous $[\text{Ni}]^{2+}$ and $[\text{Fe}]^{2+}$ complexes.^{44,53} Figure 7 highlights how slight changes in the central metal orbitals lead to substantive changes in the metal–ligand orbital interactions and on the overall outcome of reduction events. Apart from affecting the coordination number via loss of the axial solvent ligand, the molecular orbitals shown in Figure 7 show distinct differences in the type of reduction event taking place within the complexes. Indeed, in contrast to the metal-centered one-electron reduction of $[\text{Co}]^{2+}$ to $[\text{Co}]^+$ to generate a complex with a formally reduced Co^+ center bound to a neutral tpyPY2Me ligand, one-electron reductions of $[\text{Fe}]^{2+}$ and $[\text{Ni}]^{2+}$ involve a primarily ligand-centered event, resulting in $[\text{Fe}]^+$ and $[\text{Ni}]^+$ complexes with formal Fe^{2+} and Ni^{2+} centers bound to a reduced $(\text{tpyPY2Me}^{\bullet})^-$ ligand. It is noteworthy that a comparison of the first reduction events for the iron, cobalt, and nickel analogues exhibit a continuum of all the three primary outcomes for reducing a transition metal complex bound to a redox-active ligand: (1) a pure ligand-based reduction (Fe), (2) a pure metal-based reduction (Co), and (3) a ligand-based reduction with a large characteristic of coupling between a metal d orbital and a ligand π^* orbital, resulting in a broken symmetry configuration (Ni).

Despite the marked differences in electronic structure for the singly reduced forms of the iron, cobalt, and nickel tpyPY2Me complexes, all the three metals share a similar electronic characteristic upon introduction of a second reduction event. Comparing DFT results obtained for $[\text{Co}]^0$ (Figure 6, right) to those of $[\text{Fe}]^0$,⁴⁴ and $[\text{Ni}]^0$,⁵³ we observe that each first-row transition complex adopts a high-spin, unreduced formal M^{2+} center bound to a doubly reduced $(\text{tpyPY2Me}^{\bullet\bullet})^{2-}$ ligand. Thus, in the case of the second reduction event, the complexes differ only in the extent of d/ π^* orbital coupling between the metal atom and the surrounding terpyridine ligand fragment.

CONCLUDING REMARKS

In summary, we have presented a comparative study of redox behavior for first-row transition metal complexes bearing the redox-active ligand tpyPY2Me and applications of iron and cobalt congeners for photochemical CO_2RR . In particular, we established that the $[\text{Fe}(\text{tpyPY2Me})]^{2+}$ ($[\text{Fe}]^{2+}$) complex is an extremely active and selective catalyst for photochemical reduction of CO_2 to CO, achieving up to 99% CO product selectivity with the total TON and initial TOF exceeding 30,000 and 900 min^{-1} , respectively. Electrochemical analysis shows that the $[\text{Co}]^{2+}$ complex has a markedly higher thermodynamic barrier for reduction by two electrons compared to the $[\text{Fe}]^{2+}$ analogue, with a negative potential shift of 770 mV due to a large separation of the first and second reductions in the cobalt derivative. This difference in redox behavior is manifested in $[\text{Co}]^{2+}$ showing lower activity,

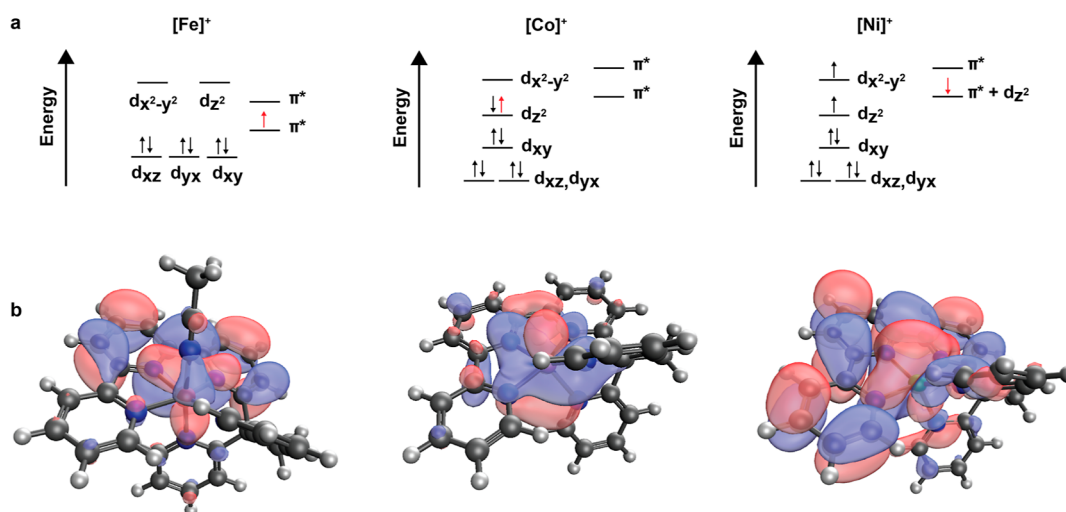


Figure 7. DFT analysis comparing varying metal–ligand redox behavior of iron, cobalt, and nickel metal centers bound to a redox-active tpyPY2Me ligand framework upon one-electron reduction. For each complex, the resulting first reduction of $[M]^{2+}$ to $[M]^+$ is shown with (a) qualitative molecular orbital diagram consistent with DFT calculations, with newly added electrons highlighted in red, and (b) relevant computed molecular orbitals. The results show that the one-electron reduction of $[Co]^{2+}$ is primarily metal-centered while the $[Fe]^{2+}$ and $[Ni]^{2+}$ analogues are ligand-centered.

selectivity, and stability for electrochemical CO_2 RR relative to $[Fe]^{2+}$, with electrode deposition and low Faradaic efficiency observed during electrolysis for the former. These reactivity trends are mirrored in photochemical CO_2 RR, where the $[Fe]^{2+}$ system is 20-fold more active than the $[Co]^{2+}$ congener under similar conditions. DLS analysis shows that the photolysis of the cobalt system leads to formation of nanoparticles with loss of CO_2 RR activity, whereas the iron system remains homogeneous, consistent with trends in electrochemical stability. Electronic structure calculations predict that the origin of the disparate redox behavior and catalysis observed for the $[Fe]^{2+}$ and $[Co]^{2+}$ complexes is the electronic configuration of the reduced species. Going from $[Co]^{2+}$ to $[Co]^+$, a metal-centered reduction is predicted, which supports the observed loss in selectivity for CO_2 RR and increased HER from a formal Co^+ species. Subsequent reduction of $[Co]^+$ to $[Co]^0$ is accompanied by a significant rearrangement of electrons to a metal–ligand coupling interaction in the form of an open-shell doublet with a formal Co^{2+} center bound to a doubly reduced (tpyPY2Me $^{••}$) $^{2-}$ ligand. Indeed, bulk electrolysis and photocatalysis experiments suggest that this open-shell doublet $[Co]^0$ species is catalytically competent for reduction of CO_2 to CO but is not a stable species, leading to inactive film or nanoparticle formation. In contrast, two-electron reduction to $[Fe]^0$, which bears an open-shell singlet electronic configuration driven by metal–ligand exchange coupling, proceeds at potentials that are nearly 800 mV more positive compared to the cobalt analogue and shows high activity for both electrochemical and photochemical CO_2 RR. This work highlights the value of metal–ligand exchange coupling as a design principle for tuning multielectron reactivity, presaging the use of this concept for advancing a broader array of catalytic transformations.

EXPERIMENTAL SECTION

Electrochemical Measurements. All electrochemical measurements were performed using a Bioanalytical Systems, Inc. (BASi), Epsilon potentiostat. CV data were measured in a three-electrode cell equipped with a GC disk (3.0 mm

diameter) working electrode, a platinum wire counter electrode, and a silver wire in a porous Vycor tip glass tube filled with the electrolyte as a pseudo-reference electrode. The final concentration of the $[Co]^{2+}$ or $[Fe]^{2+}$ catalyst was 1 mM. The working electrode was polished prior to each experiment with 0.05 μm alumina powder on a felt pad. The electrolyte was 0.1 M tetrabutylammonium hexafluorophosphate (NBu_4PF_6) in dry CH_3CN and sparged with either Ar or CO_2 as indicated. At the conclusion of each experiment, the pseudo-reference potentials were referenced against ferrocene/ferrocene ($Fc^{+/0}$) used as an external standard. The scan rates for all CVs were 100 mV/s unless otherwise noted. All voltammograms were obtained with iR compensation. CPE experiments were performed in a gastight PEEK electrolysis cell with a working compartment (12 mL liquid volume) and counter compartment (4 mL liquid volume) separated by an ultrafine glass frit. Either a 1 cm^2 GC plate (Tokai Carbon; Kanagawa, Japan) or a $\sim 1 \times 1 \times 0.6$ cm piece of 100 ppi (pores per linear inch) RVC foam (ERG Duocel; Oakland, CA) was used as the working electrode. A 2.5 cm diameter graphite planchet (Ted Pella; Redding, CA) was used as the auxiliary electrode, and an Ag wire in a Vycor tipped glass tube filled with the electrolyte was used as a pseudo-reference electrode. The working compartment was filled with 12 mL of a 1 mM solution of the catalyst dissolved in the electrolyte (0.1 M NBu_4PF_6/CH_3CN) containing either 0.1 M or 1 M phenol as a proton source. The counter electrode chamber was filled with 4 mL of a 20 mM solution of tetrabutylammonium acetate dissolved in the electrolyte. This soluble source of acetate was sacrificially oxidized to generate CO_2 and ethane, thereby preventing GC detection of solvent oxidation by-products. Both compartments were sealed to be gastight. The working compartment was sparged with CO_2 for 10 min, then sealed, and injected with 0.50 mL of ethane as a gaseous internal standard. A CV scan was collected prior to the CPE measurement to calibrate the potential. The electrolyte solution was constantly stirred during the CPE experiment with a 1 cm stir bar. At the conclusion of the experiment, the headspace was injected directly into an SRI-GC (model

#8610C) equipped with 6' Haysep D and 13X molecular sieve chromatographic columns. Two in-line detectors were used: a TCD for H₂ detection and an FID with a methanizer for CO/CO₂/C₂H₆ detection. Analytes of interest were quantified by comparing the ratios of analyte/internal standard peak integrals to a calibration curve with known amounts of analyte.

Photocatalytic CO₂ Reduction. Experiments were conducted inside a 25 mL borosilicate culture tube equipped with a stir bar, a rubber septum, and an aluminum crimped top. These reaction vessels contained 5 mL of CH₃CN, 2 μ M catalyst, 200 μ M photosensitizer, 100 mM BIH (112 mg), and 1 M phenol (470 mg). The reaction tubes were sparged with CO₂ for 10 min, followed by injection of a gaseous internal standard (0.1 mL of C₂H₆). The reactions were placed on a stir plate 13 cm from two Kessil blue LED lamps (440 nm) for 15 or 30 min at a time and maintained at ambient temperature using a fan. Analysis of the headspace by GC was conducted for product detection as described above for CPE. The samples were then re-sparged with CO₂, and the process was repeated for the following time point of an experiment.

Computational Details. DFT calculations for free energies, reduction potentials, and LOBA oxidation state analysis were performed with the Q-Chem electronic structure package.⁷² Geometry optimizations and frequency analysis were obtained using the ω B97X-D and a mixed basis set (def2-TZVP for metal Co center and def2-SVP for ligand main group elements). Solvation energies with the CH₃CN solvent used to compute reduction potentials were approximated with single-point calculations employing a C-PCM solvent model and the ω B97M-V functional with the def2-TZVP basis set for all atoms. For all reductions of [Co]²⁺, the minimum energy spin configuration was identified through an exhaustive search of viable spin states at each reduction level. The differences in adiabatic free energy for higher lying spin states are reported with respect to the minimum energy configurations in Table S8. Additional density functionals were probed, including similar range-separated hybrid functionals such as ω B97X-D and CAM-B3LYP, but all yielded substantially less agreement with the experimental values (Table S5). The ferrocenium/ferrocene couple (Fc^{+/0}) was used as an internal standard for our reported reduction potentials. The Fc^{+/0} couple was calculated with each reported density functional before standardizing to optimize potential error cancellation.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.2c02072>.

Crystallographic X-ray crystallographic data for [Co]²⁺ (CIF)

Experimental and computational details, supplemental electrochemical, photochemical and spectroscopic data, X-ray crystallography data, and DFT geometry-optimized atomic xyz coordinates (PDF)

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Notes

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NOTE ADDED AFTER ASAP PUBLICATION

This paper was published ASAP on July 1, 2022 with errors in the TOC graphic and Figure 4. The corrected version was reposted on July 6, 2022.